Buoyancy solutions and Green's functions for two Ekman numbers

(a) Buoyancy solutions, Ek = 0.25

(b) Green's functions, Ek = 0.25

(c) Buoyancy solutions, Ek = 0.025

(d) Green's functions, Ek = 0.025

depth z/H

0.0
−0.5
−1.0

−1.0
−0.5
0.0
−1
0
1
−4
−2
0

value

Exponential forcing family
Linear forcing family
b(z), b_s = 0.0
b(z), b_s = 0.1
forcing profile B(z)

ξ/H = -0.9
ξ/H = -0.6
ξ/H = -0.3
ξ/H = -0.1
ξ/H = 0.0
Ek ∂ξ G_b(z;0)